



Inspiring Excellence

Project Report

Course Code: CSE421

Section: 19

Topic: End of Experiment (Topic 4)

Submitted by:

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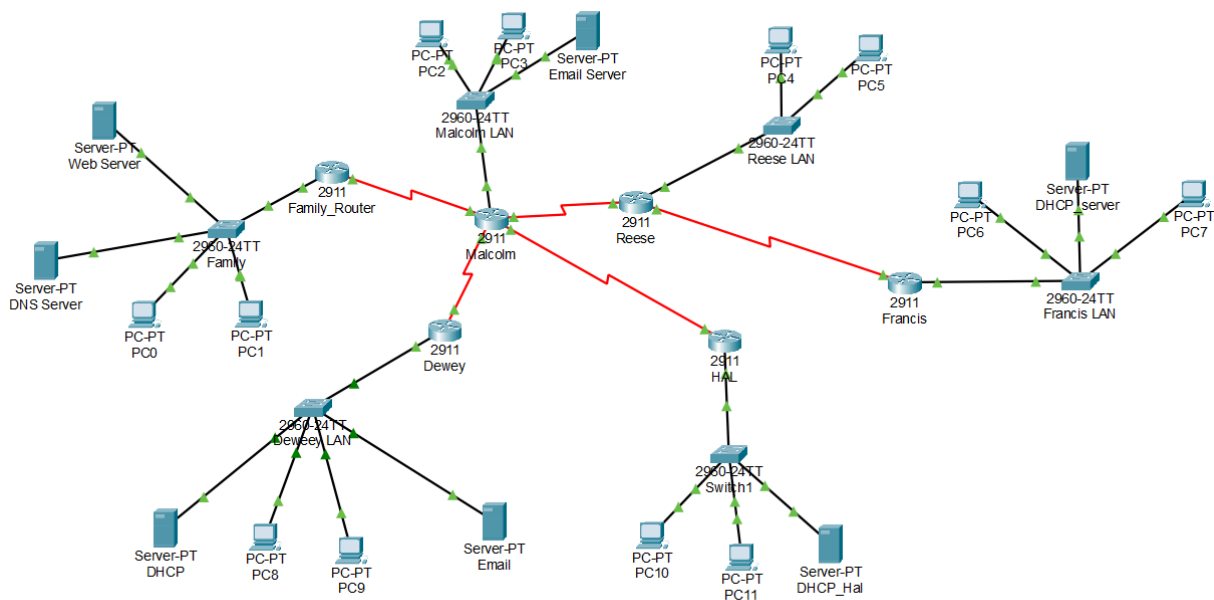
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Introduction

This report presents the design of a network topology for Malcolm and his family, providing communications between various locations, including home, school, work and other key areas. This topology has been built and implemented using Cisco Packet Tracer, with IP addressing based on Student ID: 24241333, which results in a base network address of 24.24.0.0.

1. Network Topology



2. Router Configuration Commands

A. Malcolm_Router

```
en
conf t
hostname Malcolm_Router
ip address 24.24.0.1 255.255.252.0

interface g0/0
description Malcolm School Lab LAN
ip address 24.24.0.1 255.255.252.0
no shut
exit

interface s0/3/1
description WAN to Reese Router
ip address 24.24.10.1 255.255.255.252
clock rate 64000
no shut
exit
```

```

interface s0/3/0
description WAN to HAL Router
ip address 24.24.10.9 255.255.255.252
clock rate 64000
no shut
exit

interface s0/2/0
description WAN to Family Router
ip address 24.24.10.13 255.255.255.252
clock rate 64000
no shut
exit

interface s0/2/1
description WAN to Dewey Router
ip address 24.24.10.17 255.255.255.252
no shut
exit

ip dhcp excluded-address 24.24.0.1 24.24.0.20
ip dhcp pool MALCOLM_LAN
network 24.24.0.0 255.255.252.0
default-router 24.24.0.1
dns-server 24.24.9.131
domain-name malcolm.com
exit

router rip
version 2
no auto-summary
network 24.0.0.0
passive-interface g0/0
passive-interface s0/3/0
passive-interface s0/3/1
exit

ip route 24.24.4.0 255.255.254.0 24.24.10.2
ip route 24.24.6.0 255.255.254.0 24.24.10.18
ip route 24.24.9.128 255.255.255.224 24.24.10.14

```

B. Reese_Router

```

en
conf t
hostname Reese_Router

interface g0/0
description Reese Kitchen Control Zone LAN
ip address 24.24.8.1 255.255.255.0
no shut
exit

interface s0/3/1
description WAN to Malcolm Router
ip address 24.24.10.2 255.255.255.252
no shut
exit

```

```

interface s0/3/0
description WAN to Francis Router
ip address 24.24.10.5 255.255.255.252
clock rate 64000
no shut
exit

ip dhcp excluded-address 24.24.8.1 24.24.8.20
ip dhcp pool REESE_LAN
network 24.24.8.0 255.255.255.0
default-router 24.24.8.1
dns-server 24.24.9.131
domain-name reese.com
exit

router rip
version 2
no auto-summary
network 24.0.0.0
passive-interface g0/0
passive-interface s0/3/0
exit

ip route 24.24.4.0 255.255.254.0 24.24.10.6
ip route 24.24.6.0 255.255.254.0 24.24.10.1
ip route 24.24.9.128 255.255.255.224 24.24.10.1

end
copy running-config startup-config

```

C. HAL_Router

```

en
conf t
hostname HAL_Router

interface g0/0
description HAL Workplace Office LAN
ip address 24.24.9.1 255.255.255.128
no shut
exit

interface s0/3/0
description WAN to Malcolm Router
ip address 24.24.10.10 255.255.255.252
no shut
exit

router rip
version 2
no auto-summary
network 24.0.0.0
passive-interface g0/0
exit

ip route 24.24.4.0 255.255.254.0 24.24.10.9
ip route 24.24.6.0 255.255.254.0 24.24.10.9
ip route 24.24.9.128 255.255.255.224 24.24.10.9

end
copy running-config startup-config

```

D. Francis_Router

```
en
conf t
hostname Francis_Router

interface g0/0
description Francis Military Academy LAN
ip address 24.24.4.1 255.255.254.0
no shut
exit

interface s0/3/0
description WAN to Reese Router
ip address 24.24.10.6 255.255.255.252
no shut
exit

ip route 24.24.9.128 255.255.255.224 24.24.10.5
ip route 24.24.0.0 255.255.252.0 24.24.10.5
ip route 24.24.6.0 255.255.254.0 24.24.10.5
ip route 24.24.8.0 255.255.255.0 24.24.10.5
ip route 24.24.9.0 255.255.255.128 24.24.10.5

end
copy running-config startup-config
```

E. Dewey_Router

```
en
config t
hostname Dewey_Router

interface g0/0
description Dewey Creative Experiment Lab LAN
ip address 24.24.6.1 255.255.254.0
no shut

interface s0/2/1
description WAN to Malcolm Router
ip address 24.24.10.18 255.255.255.252
clock rate 64000
no shut
exit

ip route 24.24.0.0 255.255.252.0 24.24.10.17
ip route 24.24.4.0 255.255.254.0 24.24.10.17
ip route 24.24.8.0 255.255.255.0 24.24.10.17
ip route 24.24.9.0 255.255.255.128 24.24.10.17
ip route 24.24.9.128 255.255.255.224 24.24.10.17

end
copy running-config startup-config
```

F. Family_Router

```
en
config t
hostname Family_Router

interface g0/0
```

```

description Family Home LAN
ip address 24.24.9.129 255.255.255.224
no shut
exit

interface s0/2/0
description WAN to Malcolm Router
ip address 24.24.10.14 255.255.255.252
no shut
exit

ip route 0.0.0.0 0.0.0.0 24.24.10.13
ip route 24.24.4.0 255.255.254.0 24.24.10.13
ip route 24.24.6.0 255.255.254.0 24.24.10.13 25

end
copy running-config startup-config

```

G. Verification Commands

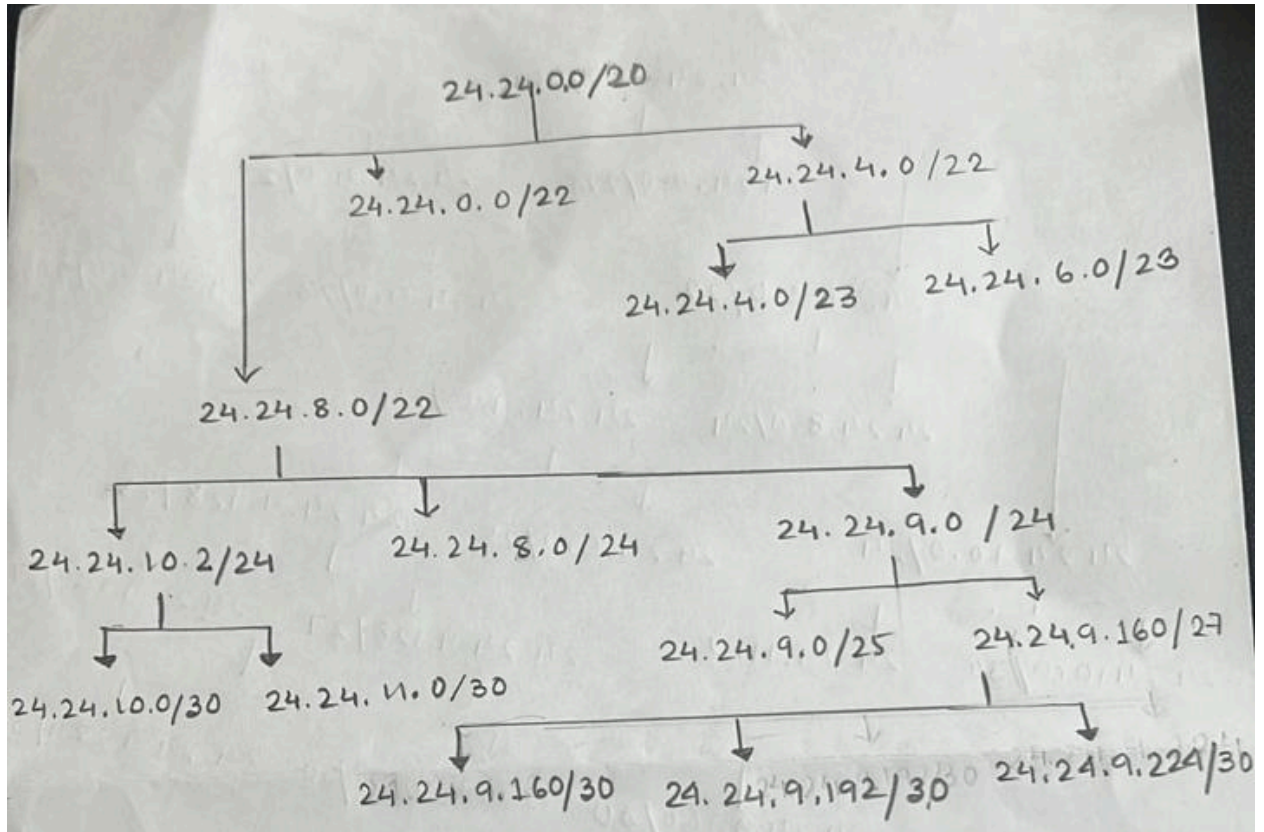
The following shows the routing table output from Malcolm_Router after full configuration:

```

Malcolm_Router>en
Malcolm_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
24.0.0.0/8 is variably subnetted, 16 subnets, 7 masks
C 24.24.0.0/22 is directly connected, GigabitEthernet0/0
L 24.24.0.1/32 is directly connected, GigabitEthernet0/0
S 24.24.4.0/23 [1/0] via 24.24.10.2
S 24.24.6.0/23 [1/0] via 24.24.10.18
R 24.24.8.0/24 [120/1] via 24.24.10.2, 00:00:03, Serial0/3/1
R 24.24.9.0/25 [120/1] via 24.24.10.10, 00:00:07, Serial0/3/0
S 24.24.9.128/27 [1/0] via 24.24.10.14
C 24.24.10.0/30 is directly connected, Serial0/3/1
L 24.24.10.1/32 is directly connected, Serial0/3/1
R 24.24.10.4/30 [120/1] via 24.24.10.2, 00:00:03, Serial0/3/1
C 24.24.10.8/30 is directly connected, Serial0/3/0
L 24.24.10.9/32 is directly connected, Serial0/3/0
C 24.24.10.12/30 is directly connected, Serial0/2/0
L 24.24.10.13/32 is directly connected, Serial0/2/0
C 24.24.10.16/30 is directly connected, Serial0/2/1
L 24.24.10.17/32 is directly connected, Serial0/2/1

```

3. VLSM Tree



4. IP Address Table

The base network address of 24.24.0.0 was constructed from Student ID 24241333, with the first two groups of digits (24 and 24) taken as the first and second octets of the network address.

LAN Subnets:

Network Name	Hosts	Subnet Mask	Network Address	First Usable IP (Gateway)	Last Usable IP	Broadcast Address
Malcolm's School	840	255.255.252.0 / 22	24.24.0.0	24.24.0.1	24.24.3.254	24.24.3.255
Francis's Academy	390	255.255.254.0 / 23	24.24.4.0	24.24.4.1	24.24.5.254	24.24.5.255
Dewey's Lab	300	255.255.254.0 / 23	24.24.6.0	24.24.6.1	24.24.7.254	24.24.7.255
Reese's Kitchen	210	255.255.255.0 / 24	24.24.8.0	24.24.8.1	24.24.8.254	24.24.8.255
Hal's Workplace	72	255.255.255.128 / 25	24.24.9.0	24.24.9.1	24.24.9.126	24.24.9.127
Family Home	25	255.255.255.224 / 27	24.24.9.128	24.24.9.129	24.24.9.158	24.24.9.159

WAN Links (Router-to-Router /30):

WAN Link	Mask	Network Address	IP Address 1 (Router A)	IP Address 2 (Router B)
WAN 1 (Malcolm ↔ Reese)	/30	24.24.10.0	24.24.10.1 (Malcolm)	24.24.10.2 (Reese)
WAN 2 (Reese ↔ Francis)	/30	24.24.10.4	24.24.10.5 (Francis)	24.24.10.6 (Reese)
WAN 3 (Malcolm ↔ Hal)	/30	24.24.10.8	24.24.10.9 (Malcolm)	24.24.10.10 (Hal)
WAN 4 (Malcolm ↔ Family)	/30	24.24.10.12	24.24.10.13 (Malcolm)	24.24.10.14 (Family)
WAN 5 (Malcolm ↔ Dewey)	/30	24.24.10.16	24.24.10.17 (Malcolm)	24.24.10.18 (Dewey)

5. Derivation of Base Network Address

The base network address for this project was calculated from Student ID 24241333 as follows:

First two digits of Student ID: 24 → First octet of base address

Next two digits of Student ID: 24 → Second octet of base address

3rd and 4th octets are set to 0

Thus the base address is: 24.24.0.0

Prefix /20 was chosen as the initial starting block, giving 4096 addresses from 24.24.0.0 to 24.24.15.255, which is enough for all the required subnets for the six network locations and WAN links.

6. Brief Notes

DHCP Configuration

Router-Based DHCP:

- In Malcolm's School Lab, the Malcolm_Router is configured as a DHCP server. The range of addresses are excluded from .1 to .20.
- Similarly, in Reese's Kitchen Control Zone, the Reese_Router interface provides the DHCP service with the same excluded range of .1-.20.

Server-Based DHCP:

- Francis's Military Academy Network employs a dedicated DHCP server in the local subnet to allocate IP addresses.
- Hal's Office Workplace Network also uses a dedicated DHCP server.
- Dewey's Creative Experiment Lab is also using a separate DHCP server within the subnet.

Routing

Routers using Static Routing:

- The Family Home Router has a default static route to send traffic to Malcolm Router.
- Standard Static Routes configured between: Francis's Military Academy Network ↔ Family Home Network.
- A floating static route has been configured between the Family Home and Dewey's Lab via Malcolm's School Lab with an AD of 25.

Routers with Dynamic Routing (RIP v2):

- The Malcolm's School Lab Router,
- Reese's Kitchen Control Zone Router, and
- Hal's Workplace Office Router are set up with RIP version 2 to dynamically share routes..

Hybrid Router (Both Static and Dynamic Routing):

The Malcolm's School Lab Router is a hybrid router. It applies dynamic routing using RIP version 2 with the Reese's Kitchen and Hal's Workplace networks (with passive interfaces applied for networks where RIP updates are not needed). It also performs static routing for networks like Francis's Military Academy, Dewey's Lab and Family Home, and redistributes these static routes into the RIP environment. As a consequence of this, the router is the central integration router of the topology where static and dynamic routing co-exist.